UNIVERSIDAD NACIONAL MAYOR DE SAN MARCOS
Doctoral Program in Deep Tech with a Focus on Artificial Intelligence and Emerging Technologies

DELIVERABLE No. 5 – REPRODUCTIBLE MACHINE LEARNING PIPELINE
Project repository (user-reported publication): https://github.com/rolandogonzales-deeptech/UNMSM-DeepTech-Deliverable-5

1. Objective and research context

The objective of this exercise is to construct and document a reproducible machine-learning workflow. The submission demonstrates version-controlled code and data, deterministic repeated execution under a fixed seed, leakage-aware model evaluation, multi-seed performance reporting, and experiment tracking. This technical exercise supports the methodological rigor required for the doctoral project on AI governance maturity in Peruvian organizations, but it is not a validation of that proposed governance model.

2. Dataset and methodology

The exercise uses the Wisconsin Breast Cancer dataset supplied through scikit-learn (569 observations). The dataset was saved as data/breast_cancer.csv. Its recorded MD5 fingerprint is 5afc23b9622f8f9ffa60824f4e97d0d5. The model is logistic regression (max_iter=5000) within a scikit-learn Pipeline with StandardScaler. Evaluation uses five-fold StratifiedKFold cross-validation with shuffling and a specified random_state. Placing scaling inside the pipeline ensures that scaling parameters are fitted on each training fold rather than on the entire dataset before cross-validation.

Reproducibility controls include Python and NumPy random seeds, PYTHONHASHSEED=0 for command-line runs, an explicit data fingerprint, pinned package versions, and a traceable Git commit identifier attached to MLflow runs. Identical outputs establish repeatability in the observed environment; they do not by themselves establish cross-platform determinism or scientific validity.

3. Repeatability verification

Measure	Run 1 (seed 42)	Run 2 (seed 42)
Accuracy mean	0.97367	0.97367
Accuracy SD across CV folds	0.01859	0.01859
Dataset MD5	Identical	Identical

Both observed executions produced the same reported values. Accuracy is expressed as a fraction; 0.97367 corresponds to 97.367%. The within-run accuracy SD is calculated across the five cross-validation folds.

4. Multi-seed experiment tracking

Five runs were recorded in the MLflow experiment session5-repro-lab, using seeds 0, 1, 2, 3, and 42. Each run records model, seed, data MD5, Git commit, cross-validation mean accuracy, and cross-validation SD. The experiment runs were linked to Git commit 0dfd8cd.

Seed	Mean accuracy	CV standard deviation
0	0.97892	0.01593
1	0.97893	0.01466
2	0.97893	0.01466
3	0.98243	0.00620
42	0.97367	0.01859

Aggregate result across seeds: 0.9786 ± 0.0031 (mean ± sample standard deviation of the five run-level mean accuracies), equivalent to approximately 97.86% ± 0.31 percentage points. This between-seed variability is distinct from the within-run cross-validation SD shown in the table.

5. Git history and deliverable artifacts

Commit	Recorded change
026a4e6	Start project: README and .gitignore
f332c67	Add dataset (breast cancer, 569 rows)
0dfd8cd	Add seeded, leakage-free training script
a9f1d69	Add results over 5 seeds
1a70d38	Add requirements and README with expected output

The five commits were confirmed in the Google Colab Git log. The local main branch was clean and configured with the GitHub origin URL before publication. The user subsequently reported completing the GitHub push; the remote publication has not been independently verified in this document.

The project includes README.md, requirements.txt, src/train.py, data/breast_cancer.csv, and results_summary.csv. The README describes the reproduction command and expected output. The package versions visible in the final notebook output were numpy==2.1.3, pandas==2.2.3, and scikit-learn==1.6.1. MLflow was used for experiment tracking, but its version was not visible in the submitted requirements output; adding an explicit MLflow version is recommended for complete environment recreation.

6. Reproduction instructions

Clone the repository, enter the project directory, install the pinned dependencies, and execute the script with the same seed:

```
git clone https://github.com/rolandogonzales-deeptech/UNMSM-DeepTech-Deliverable-5.git
cd UNMSM-DeepTech-Deliverable-5
pip install -r requirements.txt
PYTHONHASHSEED=0 python src/train.py --seed 42
```

Expected reported output includes seed 42, data MD5 5afc23b9622f8f9ffa60824f4e97d0d5, accuracy_mean 0.97367, and accuracy_sd 0.01859. Re-running the script under matching software and data conditions should reproduce the observed result. MLflow logs are stored separately from the version-controlled CSV summary.

7. Interpretation, limitations, and conclusion

This exercise provides evidence of a repeatable machine-learning pipeline and disciplined experiment reporting. It illustrates why a single high accuracy score is insufficient: preprocessing must be confined to training folds, experimental randomness must be controlled, and results should be reported across multiple seeds. The demonstrated model addresses a teaching dataset, not the thesis population; therefore, its classification accuracy must not be interpreted as evidence about AI governance maturity or compliance with Peruvian regulations.

Future adaptation to the doctoral project would replace the teaching dataset with authorized organizational governance assessments, define governance maturity labels and assessment criteria, split observations at the appropriate organizational level, and evaluate an LLM-assisted documentary assessment against independent expert judgments. These are proposed next steps, not outcomes of Deliverable 5.

Submission note: The course's primary deliverable is the GitHub repository and its reproducibility evidence. This document serves as an accompanying academic report; the repository link and published commit history should be checked before uploading to Moodle.

NB3 Thesis Adaptation — Reproducibility and Validity Assessment

Same number? The AI governance maturity classification prototype produces repeatable results under the same experimental configuration and random seed.

Valid? The experiment demonstrates a leakage-aware evaluation procedure using synthetic organizational data, but it does not establish the validity of a real-world AI governance maturity model for Peruvian organizations.

Problems? The principal methodological risks include organization-level data leakage, circular maturity labels, limited external validity, class imbalance, and insufficient independent expert validation.

Reflection on Kapoor and Narayanan (2023): Their study demonstrates that computational reproducibility alone cannot guarantee scientific validity. For my doctoral research, the AI governance maturity model and its LLM-assisted assessment component must be evaluated using independent expert judgments, organization-level data separation, transparent baselines, and reproducible experimental procedures.